

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. A white patch or a red patch — which carries the **higher** risk of becoming cancer?
2. Which salivary gland has the **highest proportion** of malignant tumours?
3. Name the commonest **benign** and commonest **malignant** salivary gland tumour.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Terminology & infections of the oral cavity

Term	Site	Term	Site
Cheilitis	Lips	Glossitis	Tongue
Gingivitis	Gum	Stomatitis	Oral mucosa

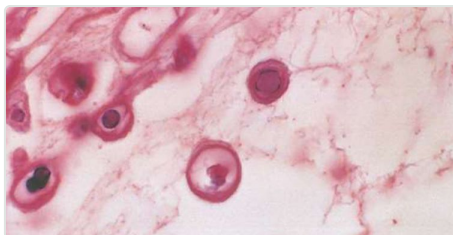
Causes of oral infection	Viral: HSV, EBV, HPV · Bacterial: Ludwig angina, diphtheria, syphilis · Fungal: candidiasis.
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2 Reactive & benign lesions — the discriminators

Lesion	Cause / trigger	The identifying feature
Recurrent herpetic stomatitis (cold sore)	Reactivated latent HSV-1 in adults (primary infection is in children, often asymptomatic).	Groups of 1-3 mm vesicles on lips, nasal orifices, buccal mucosa, gingiva, hard palate. Resolves 7-10 days (may persist if immunocompromised). Histology: glassy intranuclear inclusions .
Aphthous ulcer (canker sore)	Unknown; runs in families; up to 40% of people. Linked to coeliac disease, IBD, Behçet .	Painful single or multiple ulcers with thin fibrinopurulent exudate + erythematous halo ; heal in 7-10 days, may persist weeks.
Fibroma	Chronic irritation → reactive connective tissue hyperplasia.	Submucosal nodule on the buccal mucosa along the bite line .
Pyogenic granuloma	Unknown; possibly minor trauma or chronic irritation.	Gingiva of children, young adults, pregnancy ("pregnancy tumour"). Polypoid, richly vascular, usually ulcerated; may regress. Histology: vascular organising granulation tissue.
Mucocele	Duct blockage (retention) or rupture (extravasation — saliva leaks into stroma).	Commonest lesion of minor salivary glands ; mucous-filled cyst.
HSV reactivation triggers	Trauma · allergies · UV light · upper respiratory infection · pregnancy & menstruation · immunosuppression · temperature extremes.	
Aphthous predisposing	Stopping smoking · stress · fever · trauma · food allergy · hormonal (worse in the luteal phase, may regress in pregnancy).	
Herpes vs aphthous	Vesicles + viral inclusions = HSV. Ulcer with fibrinopurulent base and red halo , no known cause = aphthous.	



Orofacial herpetic vesicles



HSV: glassy intranuclear inclusions



Aphthous ulcer, halo



Fibroma (buccal mucosa)



Pyogenic granuloma (gingiva)



Mucocele

3 White & red patches — the discriminator

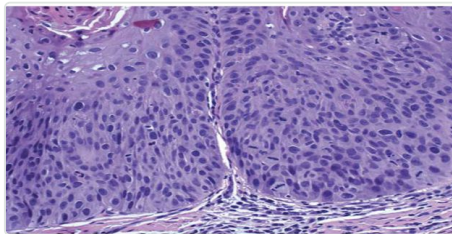
The definition does the work: **leukoplakia is a white patch that cannot be scraped off and cannot be labelled as any other disease**. White patches from obvious irritation or disease (**lichen planus, candidiasis**) are therefore **not** leukoplakia.

Both lesions: adults, typically **40-70 years, male 2:1**; multifactorial, with **tobacco** (cigarettes, pipes, cigars, chewing) the commonest risk factor.

	Leukoplakia	Erythroplakia
Appearance	White patch or plaque; cannot be scraped off	Red, velvety, possibly eroded; flat or slightly depressed
Malignant risk	3-25% (one slide says 5-25%). About 3% of the world has it	Much greater; >50%
Histology	Spectrum: increased keratinisation without dysplasia → dysplasia → invasive keratinising SCC	—



Leukoplakia, tongue



Dysplasia: pleomorphism, loss of maturation



Erythroplakia

All leukoplakia is precancerous until histology proves otherwise.

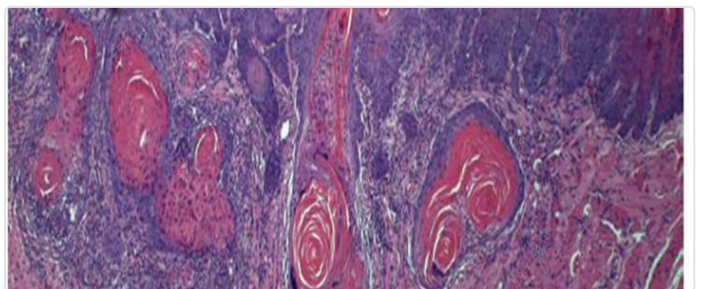
4 Oral squamous cell carcinoma — illness script

THE BLOCK'S RECURRING PATTERN

Tobacco / alcohol / HPV → **leukoplakia or erythroplakia** → dysplasia → invasive SCC. Same shape as GERD → Barrett → adenocarcinoma, H. pylori → atrophy → metaplasia → gastric cancer, and adenoma → colorectal cancer. Learn the shape once.



Oral SCC: ulceration and induration



Nests of malignant keratinocytes invading stroma

WHO gets it	Tobacco is the best-established influence (pipe smoking and smokeless tobacco especially). Alcohol is weaker alone, but the two interact to greatly increase risk . HPV 16 & 18 found in 30–50% of cases (a subset). Protracted irritation: only weakly associated.
WHY it happens	Premalignant lesions above: leukoplakia 3–25% and erythroplakia >50% transformation.
WHAT you see	Ulceration and induration of oral mucosa. Histology: nests and islands of malignant keratinocytes invading the connective tissue stroma.
WHERE it spreads	Cervical lymph nodes are the commonest regional site. Distant: mediastinal nodes, lungs, liver.
NEVER miss	95% of oral cavity cancers are SCC ; aggressive tumour with poor survival.

5 Salivary glands

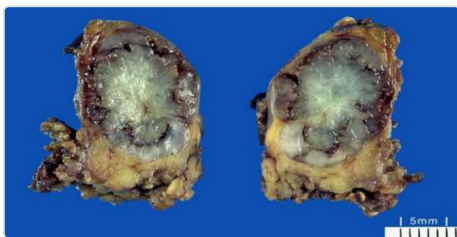
Sialadenitis	Inflammation of the gland. Trauma · viral (mumps) · bacterial (staph or strep, when a duct is obstructed by stone or food, or in dehydration) · autoimmune (Sjögren).
Mucocele	See section 2 — the commonest lesion of the minor glands.

Neoplasms — the size rule

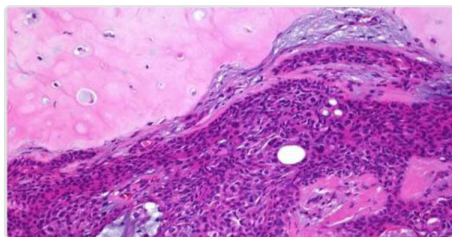
The smaller the gland, the more likely its tumour is malignant. Salivary neoplasms are only 2% of head and neck neoplasms.

Gland	Share of all SG tumours	Proportion benign
Parotid (largest)	80%	80% benign
Submandibular	15%	50% benign
Sublingual / minor (smallest)	5%	40% benign — most likely to be malignant

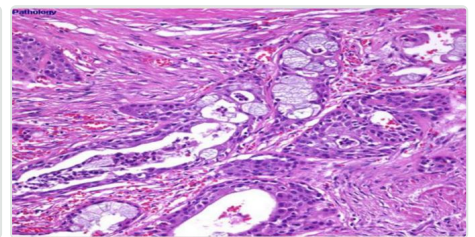
	Pleomorphic adenoma (PA)	Mucoepidermoid carcinoma (MEC)
Rank	Commonest benign tumour (50% of SG tumours; about 60% of parotid tumours)	Commonest primary malignant tumour (15%)
Also called / who	Benign mixed tumour; middle-aged females	—
Clinical	Painless, slow-growing, rounded, well demarcated	—
Composition	Ductal (epithelial) + myoepithelial cells → epithelial and mesenchymal differentiation; matrix may be myxoid, hyaline, chondroid , even osseous	Mix of squamous, mucus-secreting and intermediate cells
The catch	Recur if incompletely excised ; carcinoma arising in PA = carcinoma ex pleomorphic adenoma (malignant mixed tumour)	—



PA: well-demarcated cut surface



PA: epithelium + chondroid matrix



MEC: mixed cell populations

Full tumour list (Robbins Table 14-1, from the slide)

Benign	Malignant
Pleomorphic adenoma (50%)	Mucoepidermoid carcinoma (15%)
Warthin tumour (5%)	Acinic cell carcinoma (6%)
Oncocytoma (2%)	Adenocarcinoma NOS (6%)
Cystadenoma (2%)	Adenoid cystic carcinoma (4%)
Basal cell adenoma (2%)	Malignant mixed tumour (3%)

P Must-remember pearls

- **Red beats white for danger** — erythroplakia >50% transformation vs 3–25% for leukoplakia.
- **Leukoplakia is a diagnosis of exclusion** — lichen planus and candidiasis are not leukoplakia, and all leukoplakia is precancerous until histology says otherwise.
- **Smaller gland = more likely malignant.** Parotid 80% benign; sublingual/minor only 40%.
- **Pleomorphic adenoma recurs if incompletely excised**; MEC is the commonest malignant salivary tumour.
- **Alcohol and tobacco interact** — combined oral SCC risk far exceeds either alone; cervical nodes are the first regional site.

Q Test yourself

1. A 58-year-old pipe smoker has a red, velvety, slightly depressed patch on the floor of the mouth that cannot be scraped off. Compared with a white plaque at the same site, this lesion carries:

- A. A substantially higher risk of transformation
- B. A lower risk of malignant transformation
- C. An equal risk of malignant transformation
- D. No malignant potential
- E. A risk that cannot be compared with leukoplakia

2. Which finding would exclude a diagnosis of leukoplakia in a patient with a white oral patch?

- A. The patient uses smokeless tobacco
- B. The patch is on the buccal mucosa
- C. Histology shows keratinisation without dysplasia
- D. The patch is explained by lichen planus
- E. The patient is aged 55

3. A pregnant woman develops a richly vascular, ulcerated polypoid lesion on the gingiva. The most likely diagnosis is:

- A. Fibroma
- B. Mucocele
- C. Pyogenic granuloma
- D. Squamous cell carcinoma
- E. Aphthous ulcer

4. A 40-year-old woman has a painless, slow-growing, well-demarcated parotid mass. Histology shows ductal and myoepithelial cells in a chondroid matrix. The main risk of incomplete excision is:

- A. Conversion to Warthin tumour
- B. Distant metastasis to lung
- C. Sialadenitis
- D. Development of a mucocele
- E. Local recurrence

5. A tumour is found in a minor salivary gland of the palate. Compared with a parotid tumour it is:

- A. Less likely to be malignant
- B. More likely to be malignant
- C. Equally likely to be malignant
- D. Almost certainly a pleomorphic adenoma
- E. Impossible to compare without biopsy

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • A** — Erythroplakia carries a much greater risk of malignant transformation (>50%) than leukoplakia (3–25%).
- 2 • D** — Leukoplakia cannot be characterised as any other disease; white patches due to lichen planus or candidiasis are not leukoplakia.
- 3 • C** — Pyogenic granuloma — gingival, richly vascular granulation tissue, seen in pregnancy (“pregnancy tumour”).
- 4 • E** — Pleomorphic adenoma recurs if incompletely excised; carcinoma arising in it is called carcinoma ex pleomorphic adenoma.
- 5 • B** — The smaller the gland, the more likely its tumour is malignant: minor glands are only 40% benign vs 80% for the parotid.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam